



CPSC 100

Computational Thinking

Data Representation

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Agenda

- Course Admin
- Learning Goals
- Review of Binary/Ternary
- Hexadecimal
- RGB Colours

Course Admin

Learning Goals



Learning Goals

After this **today's lecture**, you should be able to:

- Recognize **hexadecimal numbers** and their role in data representation.
- Count in different **standard number bases** (i.e. 2, 10, 16).
- **Translate** numbers between binary, hexadecimal, and decimal without a calculator

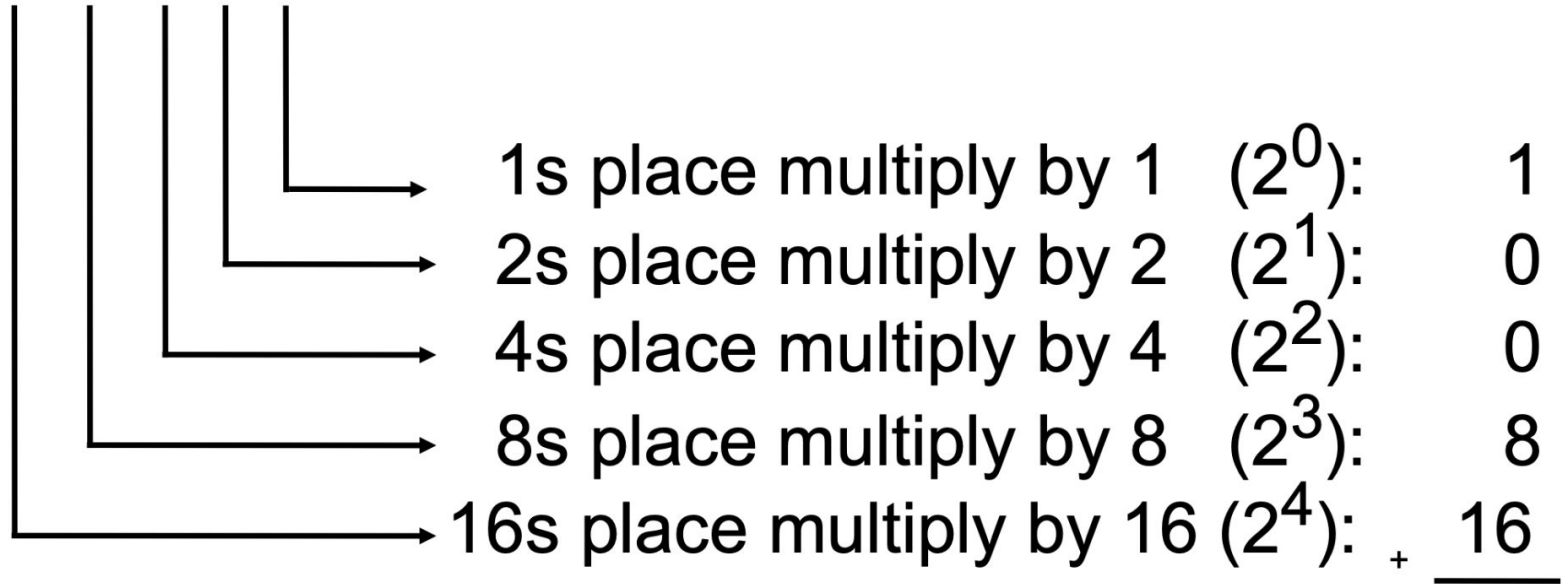
After watching the **take home-video**, you should be able to:

- Recognize the difference between binary and ternary numbers
- Translate numbers between binary, hexadecimal, decimal and **ternary**

Convert Binary to Decimal

Recall

0b 1 1 0 0 1



Total in decimal (add them up): 25

Algorithm to Convert Decimal to Binary

- Start with a decimal number n (0 to 255)
 - Since 255 is the maximum, we need 8 bits.
- Check the highest power of 2 that fits into n :
 - If $n < 128$, set the 1st (leftmost) bit to 0; otherwise, set it to 1 and subtract 128 from n ($n=n-128$)
 - If $n < 64$, set the 2nd bit to 0; otherwise, set it to 1 and subtract 64 from n . ($n=n-64$)
 - If $n < 32$, set the 3rd bit to 0; otherwise, set it to 1 and subtract 32 from n . ($n=n-32$)
 - Continue this process for 16, 8, 4, 2, and finally 1.
- By the end, n will be either 0 or 1, which is the last (8th) bit.

Algorithm to Convert 217 to Binary

- $n = 217$
- Check the highest power of 2 that fits into n :
 - $\times 217 < 128 (2^7)$, 1st Bit = **1**
 - $n = 217 - 128 \Rightarrow 89$
 - $\times 89 < 64 (2^6)$, 2nd bit = **1**
 - $n = 89 - 64 \Rightarrow 25$
 - $\checkmark 25 < 32 (2^5)$, 3rd bit = **0**
 - *No subtraction required*
 - $\times 25 < 16 (2^4)$, 4th bit = **1**
 - $n = 25 - 16 \Rightarrow 9$
 - $\times 9 < 8 (2^3)$, 5th Bit = **1**
 - $n = 9 - 8 \Rightarrow 1$
 - $\checkmark 1 < 4 (2^2)$, 6th bit = **0**
 - *No subtraction required*
 - $\checkmark 1 < 2 (2^1)$, 7th bit = **0**
 - *No subtraction required*
 - **Final bit = $n = 1 (2^0) \rightarrow$ 8th bit = **1****
- **Final Answer: 11011001**

Recap



Q: What is the binary number for 9?



- A. 10001
- B. 01010
- C. 01011
- D. 01000
- E. 01001



Hexadecimal Numbers



Hexadecimal Numbers

- Computers don't understand Decimal (base 10)
- Humans have trouble with Binary (base 2)
 - 1111111110011000111000101010
 - Writing so many 0's and 1's is tedious and error prone
- **Hex digits**, short for hexadecimal (base 16)
 - Compromise between humans and computers
 - **Easy to convert between Hex and Binary**



Hexadecimal Numbers

- The digits of the hexadecimal numbering system are
 - 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F
- Because there are 16 digits, they can be represented perfectly by the 16 symbols of 4-bit sequences:
 - The bit sequence **0000** is **hex 0** Bit sequence **0001** is **hex 1**
 - Bit sequence 1111, is hex F
 - Sometimes we use the representation **0x[number]** to show that a number is in **hex**.



Hexadecimal Numbers to Binary

- Because each hex digit corresponds to a 4-binary sequence, it's easy to translate between hex and binary
 - Hex → binary: replace each hexadecimal digit by the four corresponding binary digits in the conversion table

Hex:

F

A

B

4

Binary:

1111

1010

1011

0100



Binary to Hexadecimal

- Binary → hex:
 - Put extra 0's at the left of the binary number as necessary so that the total number of digits is a multiple of 4
 - then reverse the hex → binary conversion process

Binary:	0010	1011	1010	1101
Hex:	2	B	A	D



Activity - Fill in the Gaps

(Take Home)

Decimal	Binary	Hex
00	000	0
01	001	1
02	010	
03		
04	100	
05		
06		6
07		

Decimal	Binary	Hex
08		
09	1001	
10		A
11	1011	
12		
13	1101	
14		
15		F

(leading 0's, shown in gray, are useful for some conversions)



Q: Translate 0xAFF1 to binary

- A. 1111 1101 1101 0001
- B. 1010 1101 1101 0001
- C. 0001 1111 1111 1010
- D. 1111 1010 1111 0001
- E. 1010 1111 1111 0001



Learning Goals

After this **today's lecture**, you should be able to:

- Understand how colours are represented in the RGB model
- Describe how RGB colours are stored and represented in computing.
 - Recognize that each colour (R, G, B) has 256 intensity levels (0-255).
- Convert between different RGB representations: Decimal, Binary, & Hex

After watching the **take-home video**, you should be able to:

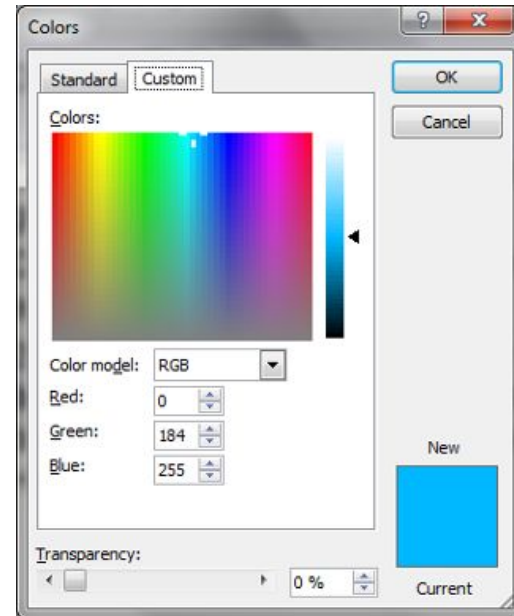
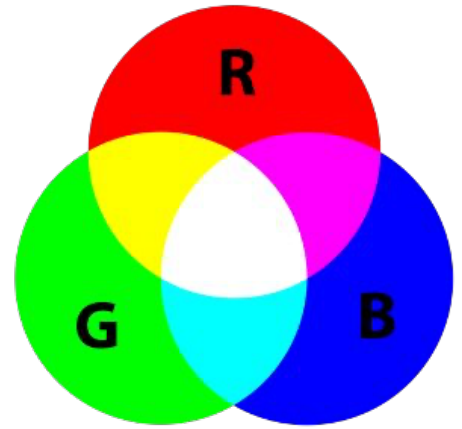
- Analyze how RGB colours mix to produce new colours.
- Explain and apply colour theory concepts to digital designs

RGB Colours



RGB Colours

- Monitors, phone screens, and TVs make different colours by mixing **Red**, **Green**, and **Blue** lights
- Computer applications use 256 intensities (8 bits) for each of red, green, and blue.





RGB Colours

Black is the absence of light:

- **0000 0000 0000 0000 0000 0000** (Binary)
- **0 0 0 0 0 0** (Hex)
 - RGB bit assignment for black

White is the full intensity of each color:

- **1111 1111 1111 1111 1111 1111** (Binary)
- **F F F F F F** (Hex)
 - RGB bit assignment for white

- https://www.w3schools.com/colors/colors_picker



Red: #FF0000



Green: #00FF00



Blue: #0000FF



Black: #000000

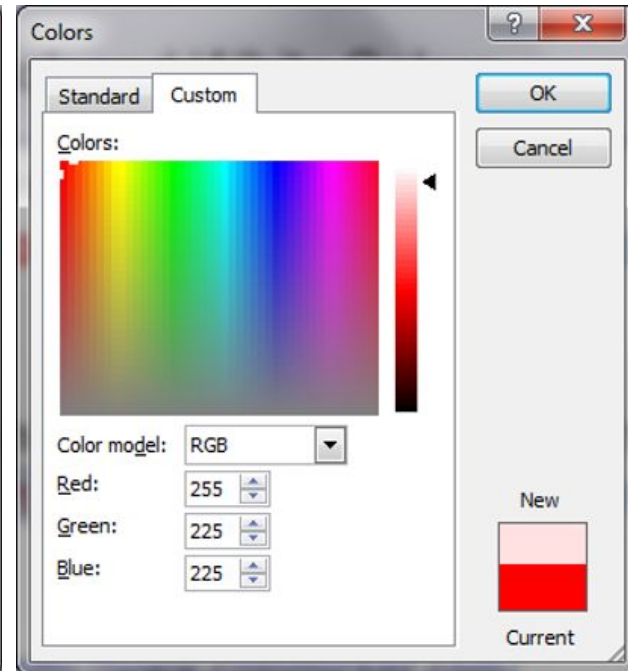
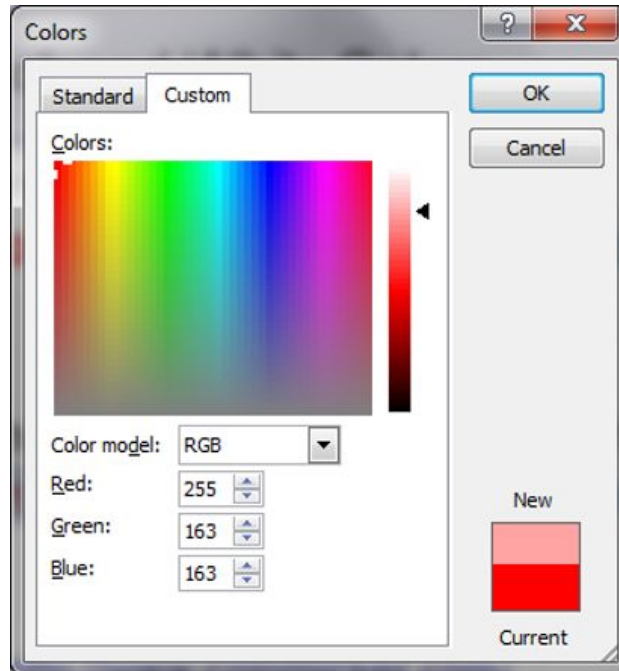
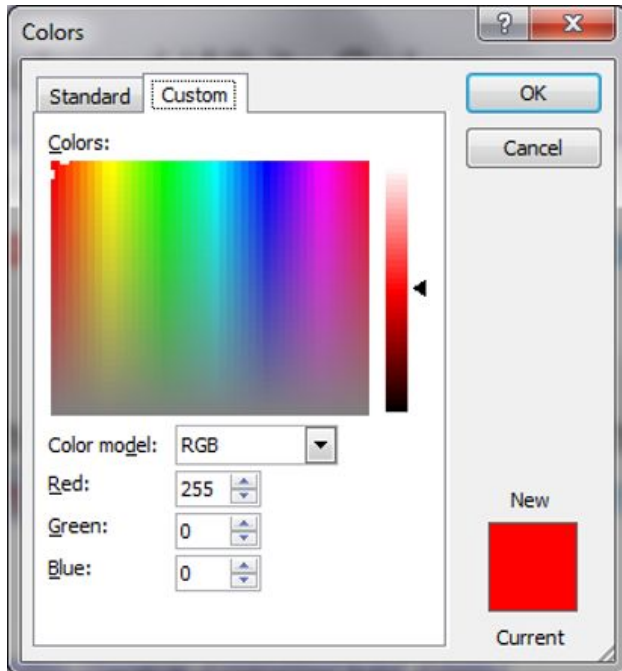


White: #FFFFFF



Grey: #CCCCCC

RGB Colours in Decimal





RGB Colours

• Colour	Decimal	Hex
• Red	→ (255, 0, 0) →	#FF0000
• Green	→ (0, 255, 0) →	#00FF00
• Blue	→ (0, 0, 255) →	#0000FF
• White	→ (255, 255, 255) →	#FFFFFF
• Black	→ (0, 0, 0) →	#000000



RGB Colours

- **Colour** **Decimal** **Hex**
- **R**ed → (255, 0, 0) → #FF0000
- **G**reen → (0, 255, 0) → #00FF00
- **B**lue → (0, 0, 255) → #0000FF
- White → (255, 255, 255) → #FFFFFF
- Black → (0, 0, 0) → #000000

Recall: Hexadecimal is a base-16 system where:

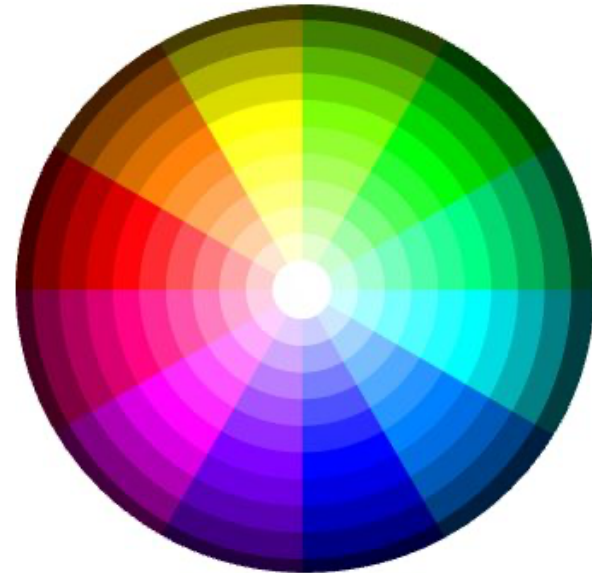
- Digits 0-9 represent values 0-9.
- Letters A-F represent values 10-15.
- Example: FF (Hex) = $(F \times 16^1) + (F \times 16^0)$
 $(15 \times 16) + (15 \times 1) \Rightarrow 255$

Decimal	Hex
0	0
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9
10	A
11	B
12	C
13	D
14	E
15	F



Q: Which colour best describes the one represented by the hexadecimal colour code: #32CD32?

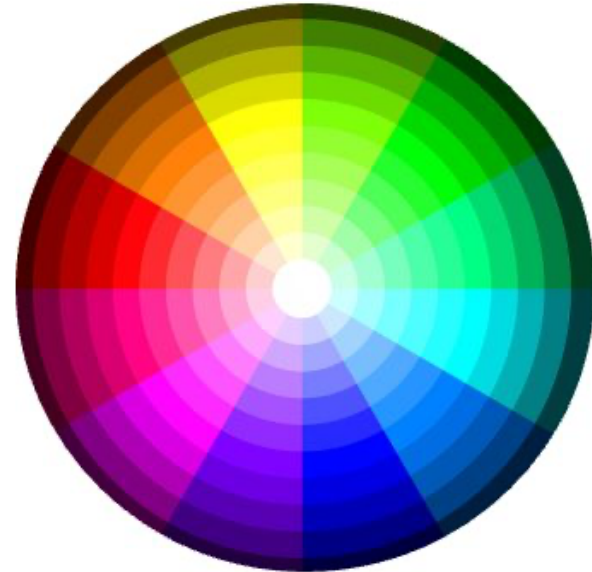
- A. Shade of red
- B. Shade of blue
- C. Shade of green
- D. Shade of purple
- E. Shade of yellow





Q: Which colour best describes the one represented by the hexadecimal colour code: #800B80?

- A. Shade of red
- B. Shade of blue
- C. Shade of green
- D. Shade of purple
- E. Shade of yellow



Mini-Activity



Color Mixing: Match the Colour to its Hex Rep.

Hex. Rep.	shade of ...
#FFA933	Yellow
#FF99FF	Pink or Magenta
#EAE51D	Blue
#A1A2A3	Orange
#1234F8	Grey



Color Mixing: Match the Colour to its Hex Rep.

Hex. Rep.		shade of ...
#FFA933		Yellow
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#1234F8		Grey



Take-Home Practice

Convert these numbers to binary

- 16
- 27
- 30
- 58
- 98
- 78
- 100
- 9
- 34

**Convert 1100 1010 1111 1110
to hexadecimal**

Convert 0xA19C to decimal

Convert 48 to hexadecimal

Take Home Video

Colour Theory

https://youtu.be/_2LLXnUdUlc?si=ZC0gCVCkhlmnc3KT



Another Useful Video

https://youtu.be/CBYhwc4WSI?si=fITtLe_y5ZyqDQTA&t=57



Wrap up